Contents

- Introduction 1
2. Data Wrangling Process 2
- 2.1 Data Loading and Discovery 2
- 2.2 Data Structuring 2
2.3 Data Cleaning 2
- 2.4 Data Validation 3
3. Code Explanation 3
- 3.1 Data Loading and Discovery Code 3
- 3.2 Data Structuring Code 4
- 3.3 Data Cleaning Code 4
- 3.4 Data Validation Code 4
4. Results and Discussion 4
5. Conclusion 4
6. References ..... **Error! Bookmark not defined.**

- Introduction

Cybershujaa program

Week 2 assignment

Netflix data wrangling

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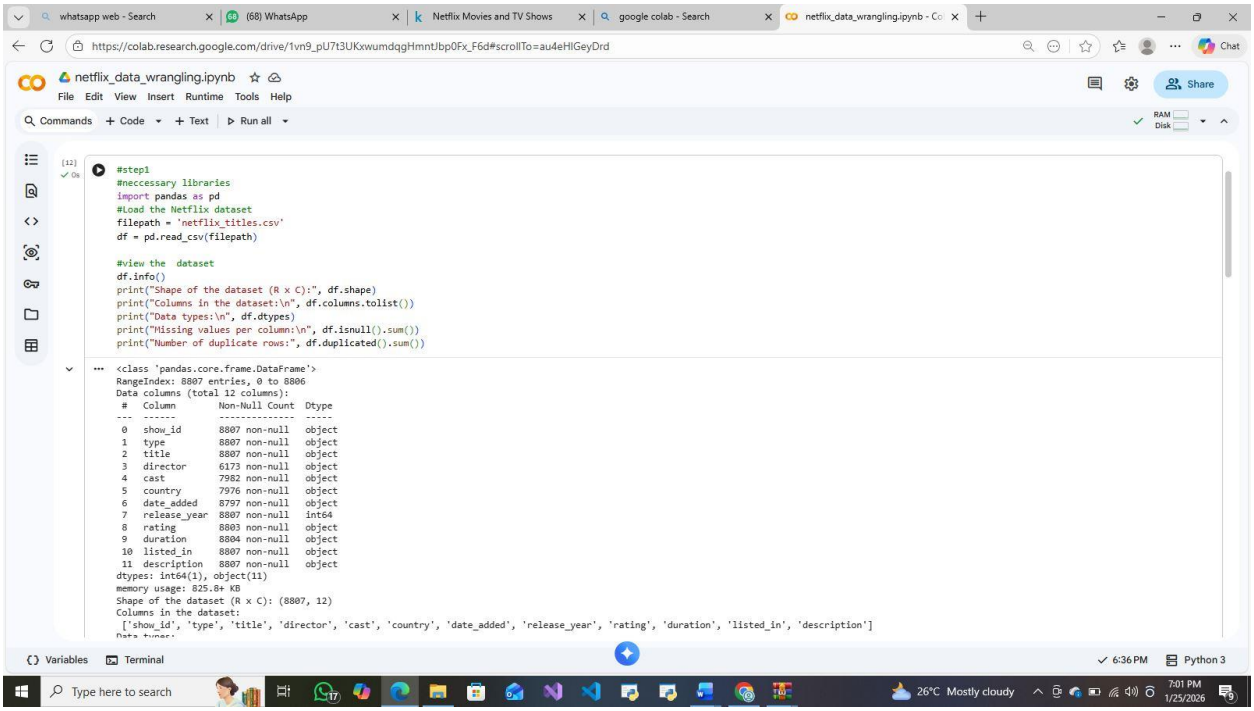
2. Data Wrangling Process

- 2.1 Data Loading and Discovery



- 2.2 Data Structuring

```
[22]
#step2-structuring
# Convert date_added to datetime
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed')

# Split duration into value and unit
df[['duration_value', 'duration_unit']] = df['duration'].str.extract(r'(\d+)\s*(\w+)')
df['duration_value'] = pd.to_numeric(df['duration_value'])
```

2.3 Data Cleaning

```
25]
#step 3-cleaning the data
#Handle duplicates and missing values
df = df.drop_duplicates()
if 'description' in df.columns:
    df = df.drop(columns=['description'])
df['director'] = df['director'].fillna('Not Given')
df['cast'] = df['cast'].fillna('Not Given')
df['country'] = df['country'].fillna('Not Given')
df = df.dropna(subset=['date_added', 'rating', 'duration'])
```

- 2.4 Data Validation

```
#step 4-validation
print(df.sample(5))
df = df.reset_index(drop=True)
```

```
     show_id     type                        title  \
8130   s8147    Movie          Take Me Home Tonight
2632   s2633    Movie  No Direction Home: Bob Dylan
7689   s7705    Movie           Paranormal Activity
3674   s3675  TV Show  PILI Fantasy: War of Dragons
6924   s6935    Movie     Harry Benson: Shoot First

                       director  \
8130              Michael Dowse
2632            Martin Scorsese
7689                  Oren Peli
3674                  Not Given
6924  Justin Bare, Matthew Miele
```

es Terminal

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3. Code Explanation

objectives

The objectives of the assignment are to:

- Load the Netflix dataset from a CSV file and explore its structure using pandas.
- Perform data discovery to assess data types, missing values, and quality issues.
- Clean the dataset by handling duplicates, missing values, and formatting inconsistencies.
- Transform and enrich the dataset using techniques like filtering, sorting, grouping, and feature extraction.
- Validate the final dataset by checking consistency, completeness, and logical accuracy.
- Export the final cleaned dataset to a .csv file ready for analysis or visualization

- 3.1 Data Loading and Discovery Code

The Netflix dataset was loaded from Kaggle and explored using pandas. The dataset contains information about Netflix shows, including title, director, cast, country, release year, rating, and duration.

- 3.2 Data Structuring Code

The date_added column was converted to datetime, and the duration column was split into duration_value and duration_unit.

- 3.3 Data Cleaning Code

Duplicate rows were removed, and missing values were imputed with 'Not Given' or dropped if necessary. The description column was dropped as it was not needed for analysis.

- 3.4 Data Validation Code

The cleaned data was validated by checking for consistency and completeness.

4. Results and Discussion

Link to notebook:
https://colab.research.google.com/drive/1vn9_pU7t3UKxwumdqgHmntJbp0Fx_F6d?usp=sharing

5. Conclusion

This project demonstrated the importance of data wrangling in preparing datasets for analysis. The cleaned dataset can now be used for further analysis or visualization.